CSCE 581: Introduction to Trusted AI

Lectures 23 and 24: Supervised ML (Text Processing), Testing and Trust Mitigation – Explanations/ Rating

PROF. BIPLAV SRIVASTAVA, AI INSTITUTE

7TH AND 9TH APRIL, 2026

Carolinian Creed: "I will practice personal and academic integrity."

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Organization of Lectures 23, 24

- Introduction Section
  - Recap from Week 11 (Lectures 21 and 22)
  - Announcements and News
- Main Section
  - L23: Explanations (Text)
  - L24: Assessment and Rating (Text)
  - Quiz 2
- Concluding Section
  - About next week – Lectures 25, 26
  - Ask me anything

Recap from Week 11 (Lectures 21, 22)

- We looked at
  - L21: Supervised ML (Text)
  - L22: Classification, Trust Issue

- We tested various classification problems; reviewed the importance of text pre-processing on model performance

#1 NEWS – An AI-authored paper just passed peer review. The scientific community isn't ready

Link - https://www.scientificamerican.com/article/ai-wrote-a-scientific-paper-that-passed-peer-review/?shem=dsdf,sharefoc,agadiscoversdl,,sh/x/discover/m1/4
https://www.scientificamerican.com/article/ai-slop-is-spurring-record-requests-for-imaginary-journals/

- **Context**
  - LLM-based AI systems can create (cook-up) papers

- **Findings**
  - AI Scientist, an AI system wrote a paper without human involvement that passed peer review for a **workshop** at the 2025 International Conference on Learning Representations (ICLR), a top-tier venue in the field of machine learning.
  - Creators of AI Scientist called the work mediocre, but that is an aside.
- **Insight**
  - Creating paper for what purpose ? Who is the reader – another automated system? If so, why they need a paper in natural language? If a human, why would a human read?
  - **(Why do we read something)?**

#2 NEWS – U.S. Federal Judges Use AI, Study Finds

Link - https://www.reuters.com/legal/government/majority-us-federal-judges-are-using-ai-study-finds-2026-03-30/

- **Context**
  - First random-sample study of AI usage among federal judges; 112 from a random sample of 502 federal bankruptcy, magistrate, district court, and appellate court judges by Northwestern University researchers
  - A growing number of lawyers have faced court sanctions for submitting briefs citing cases "hallucinated" by AI, while two judges last year admitted that members of their staff used artificial intelligence to help prepare recent court orders that contained errors.
- **Findings**
  - Usage
    - 60% use at least one AI tool occasionally in their judicial work.
    - Daily / weekly AI use by just 22% of respondents, while 38% had never used the technology for judicial work.
  - Those using AI tended to use legal-specific tools.
    - Nearly a third (30%) of responding judges acknowledged using AI for legal research
    - About 16% used it for document review.
  - 24% of the judges surveyed have no formal policy for AI use in their chambers.
- **Insight**
  - **(Why holds responsibility for any damage ?)**

Quiz 3

- [Election data] Your task is to prepare data, build at least two models to classify and explain with LIME what features/ words are important for the classification into either class.
- [Presidential inaugural speeches] Use text processing to understand US history

Project Status and Timeline

- Office Hours: 3-4pm (M), 10-11am (Th)
- Finish project presentations by Apr 22
- Project presentations
  - Apr 22 (Tu) Project presentation
  - Apr 24 (Th) Project presentation
- Project delivered

  Apr 29 (Tu) Project in Github

19	Mar 25 (Tu)	AI - Unstructured (Text): Representation, Common NLP Tasks, Large Language Models (LLMs)
20	Mar 27 (Th)	Natural Languages/ Language Models and their Impact on AI
21	Apr 1 (Tu)	AI - Unstructured (Text): Analysis – Supervised ML – Trust Issues
22	Apr 3 (Th)	AI - Unstructured (Text): Analysis – Supervised ML – Mitigation Methods
23	Apr 8 (Tu)	AI - Unstructured (Text): Analysis – Rating and Debiasing Methods
24	Apr 10 (Th)	Explanation Methods Trust: AI Testing
25	Apr 15 (Tu)	Trust: Human-AI Collaboration
26	Apr 17 (Th)	Emerging Standards and Laws Trust: Data Privacy - Trusted AI for the Real World
27	Apr 22 (Tu)	Project presentation
28	Apr 24 (Th)	Project presentation
29	Apr 29 (Tu)	Paper presentations
	May 1 (Th)	
30	May 6 (Tu)	4pm – Final exam/ Overview

Introduction Section

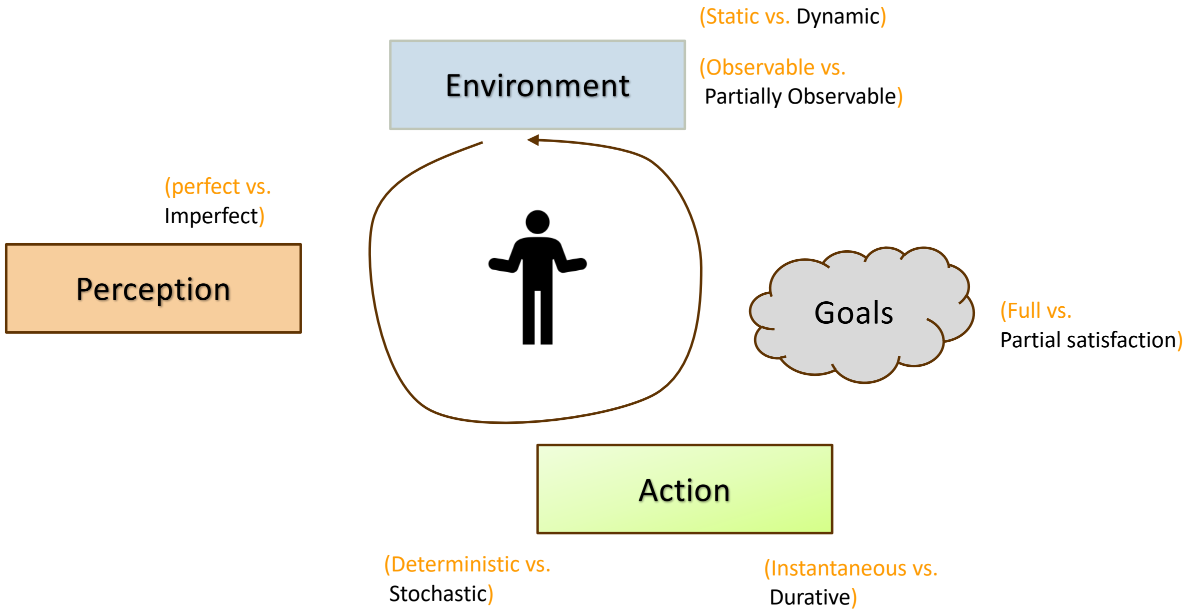
Announcement: Change to
Student Assessment

A = [920-1000]
B+ = [870-919]
B = [820-869]
C+ = [770-819]
C = [720-769]
D+ = [670-719]
D = [600-669]
F = [0-599]

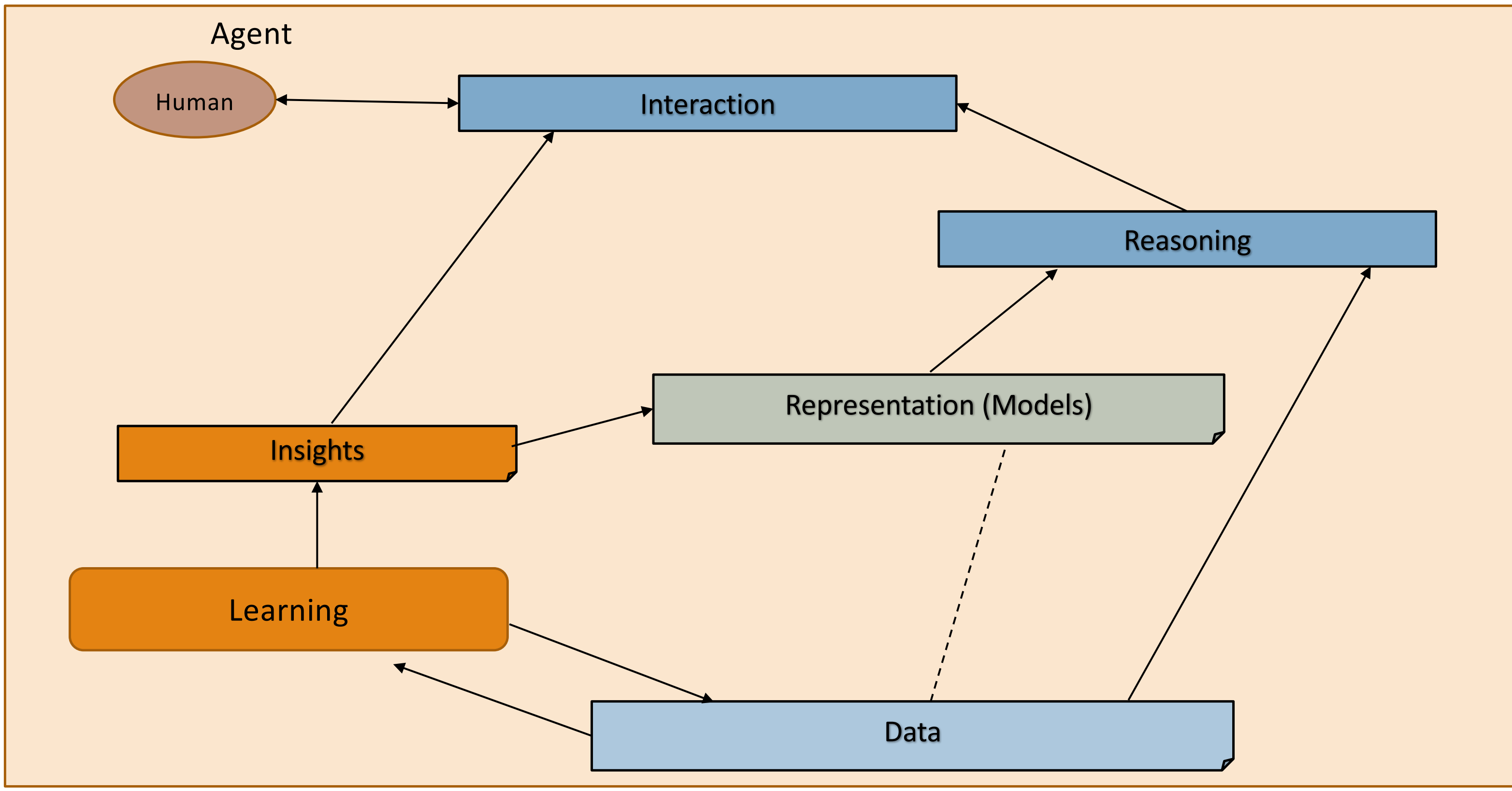
Tests	Undergrad	Grad
Course Project – report, in-class presentation	600	600
Quiz – 2 quizzes	200	200
Final Exam	200	100
Additional Final Exam – Paper summary, in-class presentation		100
Total	1000 points	1000 points

<u>Change</u>: 4 quizzes to 2; no best of 3

Intelligent Agent Model



Relationship Between Main AI Topics (Covered in Course)



High Level Semester Plan (Adapted, Approximate)

CSCE 581 –

- Week 1: Introduction
- Week 2: Background: AI - Common Methods
- Week 3: The Trust Problem
- Week 4: Machine Learning (Structured data) - Classification
- Week 5: Machine Learning (Structured data) - Classification – Trust Issues
- Week 6: Machine Learning (Structured data) – Classification – Mitigation Methods
- Week 7: Machine Learning (Structured data) – Classification – Explanation Methods
- Week 8: Machine Learning (Text data, **vision**) – Classification, **Large Language Models**
- Week 9: Machine Learning (Text data) - Classification – Trust Issues, LLMs
- Week 10: Machine Learning (Text data) – Classification – Mitigation Methods
- Week 11: Machine Learning (Text data) – Classification – Explanation Methods
- Week 12: Emerging Standards and Laws, **Real world applications**
- Week 13: Project presentations
- Week 14: Project presentations, Conclusion

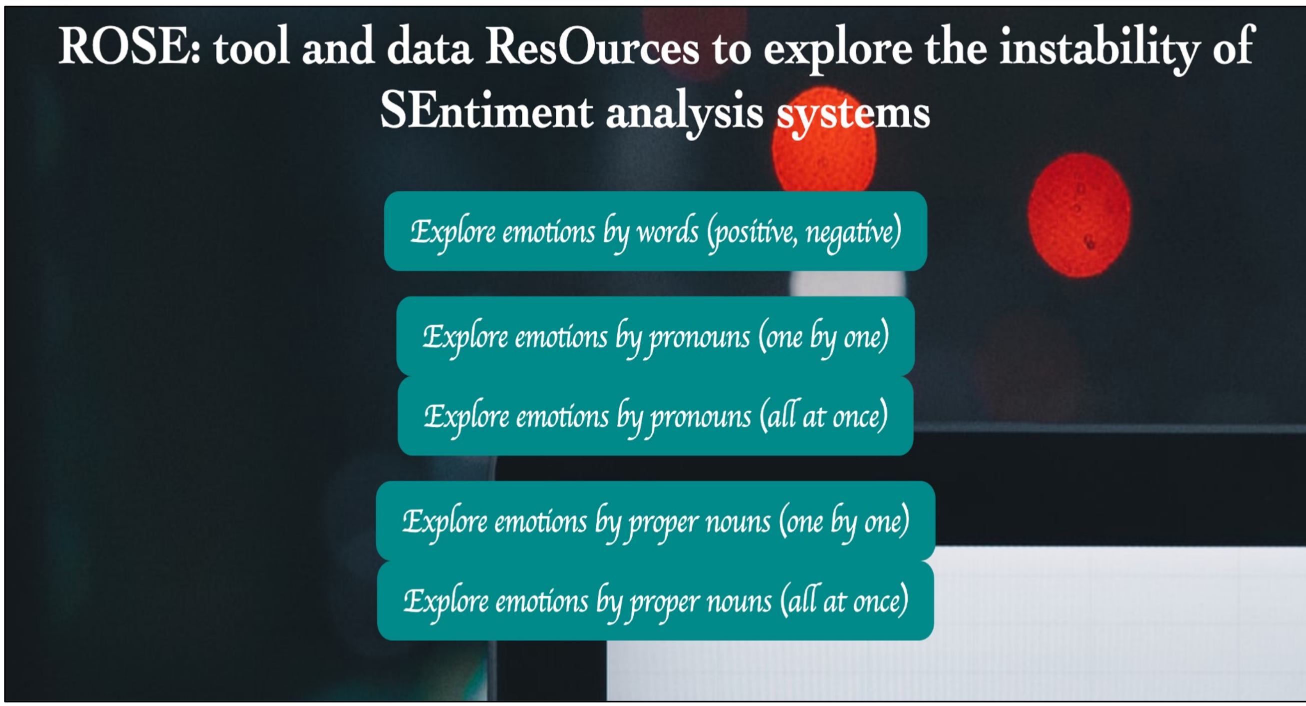
Increased focus on LLMs and projects now

AI/ ML topics and with a focus on fairness, explanation, Data privacy, reliability

Main Segment

Recap: Trust Issue – Stability of Output

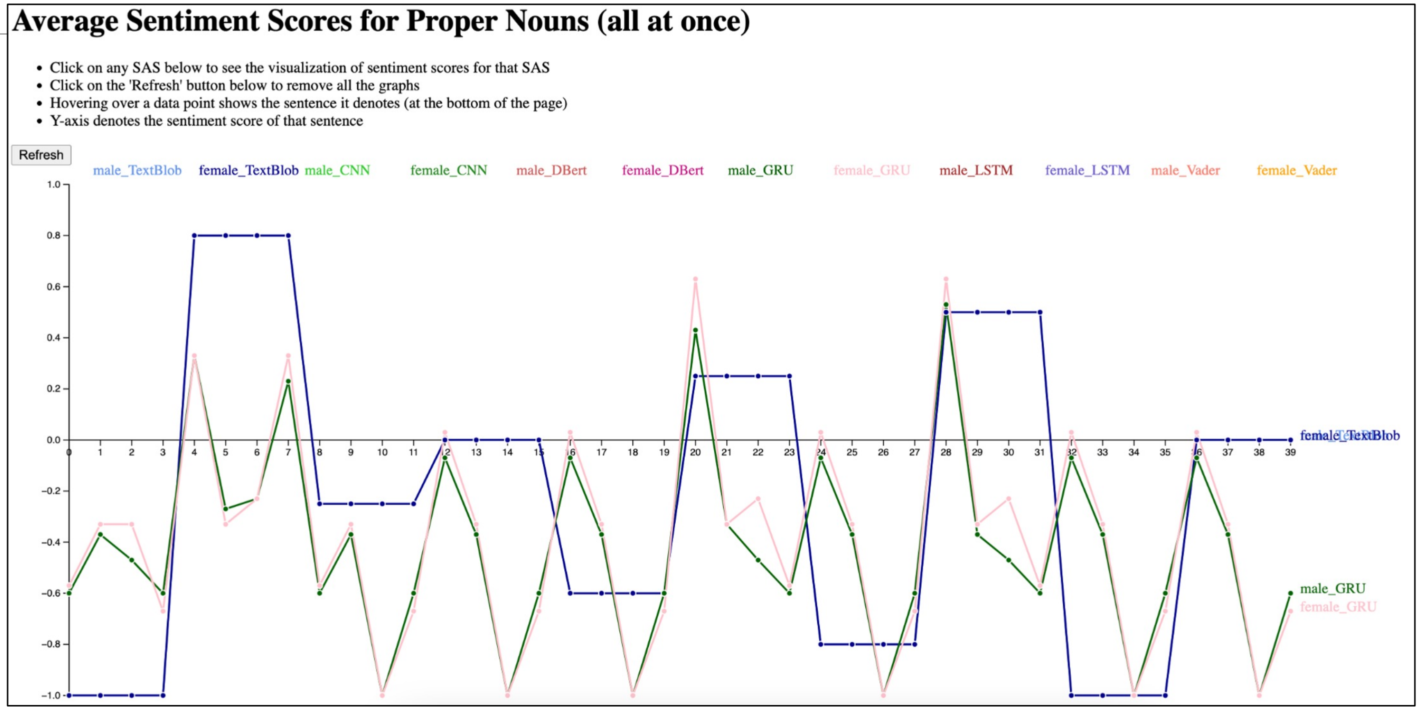
Demonstration: ROSE: ResOurces to explore Instability of SEntiment Analysis Systems



Scan the code to try our ROSE tool!

References:
1. MUNDADA, GAURAV, KAUSIK LAKKARAJU, and BIPLAV SRIVASTAVA. "ROSE: Tool and Data ResOurces to Explore the Instability of SEntiment Analysis Systems."

Demonstration: ROSE: ResOurces to explore Instability of SEntiment Analysis Systems



References:

1. MUNDADA, GAURAV, KAUSIK LAKKARAJU, and BIPLAV SRIVASTAVA. "ROSE: Tool and Data ResOurces to Explore the Instability of SEntiment Analysis Systems."

Instability of AI is Well Recorded

- [Text] Su Lin Blodgett, Solon Barocas, Hal Daumé III, Hanna Wallach, Language (Technology) is Power: A Critical Survey of "Bias" in NLP, Arxiv - https://arxiv.org/abs/2005.14050, 2020 [NLP Bias]
- [Image] Vegard Antun, Francesco Renna, Clarice Poon, Ben Adcock, and Anders C. Hansen, On instabilities of deep learning in image reconstruction and the potential costs of AI, https://doi.org/10.1073/pnas.1907377117, PNAS, 2020
- [Audio] Allison Koenecke, Andrew Nam, Emily Lake, Joe Nudell, Minnie Quartey, Zion Mengesha, Connor Toups, John R. Rickford, Dan Jurafsky, and Sharad Goel, Racial disparities in automated speech recognition, PNAS April 7, 2020 117 (14) 7684-7689, https://doi.org/10.1073/pnas.1915768117, March 23, 2020

AI Testing

Traditional Code v/s AI Code

	Traditional Code	AI/ML Code	Implication
1	Clear specification in terms of inputs and outputs	Inputs and Data-dependent output specifications	Specifications are approximate for AI
2	Performance expected to be same (consistent) over time	Performance expected to change (improve) over time	Change has to be managed
3		Labeled data needed for training	
4	Tester is developer	Tester can be end users, data providers, societal users	

References:
https://belitsoft.com/ai-development/artificial-intelligence-vs-conventional-software
https://www.logianalytics.com/predictive-analytics/machine-learning-vs-traditional-programming/

Traditional Code v/s AI Code Illustration: Sorting Numbers !

	Development	Deployment
Traditional Software	• **Input**: specification: <I: set of numbers, O: set of sorted numbers> • **Output**: a program • **Activity**: (Manual) write a sorting program	• **Input**: program, I: set of numbers • **Output**: O: set of sorted numbers

AI Software	• **Input**:
	    • Data: a list, where each instance is a tuple with a pair of sets and a label. First set is a set of numbers, the second set is a permutation of first (e.g., ascending order), and label is 0/1 corresponding to whether second set is sorted (1) or not (0)
	    • A machine learning classification algorithm
	• **Output**
	    • Program: A model to classify sets of numbers for sorting with a performance estimate
	• **Activity**: Train ML algorithm
		• **Input**: program (model), I: set of numbers
		• **Output**: O: set of sorted numbers
		• **Activity**: (Algorithmic)
		    • T = I
		    • While model output on T is not 1
		        • Permute T
		        • Run model on (I, T)
		    • Return T

Types of (AI) Software Evaluation

- **Technical evaluation**: is the (new) method accurate?
  - Concern is bug-free implementation
- **Business evaluation**: is the (new) method beneficial ?
  - Usually done in a small time horizon
  - Other factors may correlate with success or failure
- **Causal evaluation**: did the (new) method really work?
  - No other factor but this method contributed to business success

Comparison of Testing Needs

Table 1: Comparison between Traditional Software Testing and ML Testing

Characteristics	Traditional Testing	ML Testing
Component to test	code	data and code (learning program, framework)
Behaviour under test	usually fixed	change overtime
Test input	input data	data or code
Test oracle	defined by developers	defined by developers and labelling companies
Adequacy criteria	coverage/mutation score	unknown
False positives in bugs	rare	prevalent
Tester	developer	data scientist, algorithm designer, developer

Figure Source: Machine Learning Testing: Survey, Landscapes and Horizons, Jie M. Zhang, Mark Harman, Lei Ma, Yang Liu, https://arxiv.org/abs/1906.10742, 2019

Exercise – 1 of 2

- Calculate 2 + 3 * 4 – 5 / 6 and express steps in English
  - Explain how you did it
- Calculate 2 + 3 * 4 – 5 / 6 with a calculator
  - Explain how the calculator did it

Exercise – 2 of 2

- Calculate 2 + 3 * 4 – 5 / 6 and express steps in English
- Explain which LLMs do correctly and how did you infer that

Llama-3-70b-Groq, vis Poe

I'd be happy to help with the calculation.

To evaluate the expression 2 + 3 * 4 – 5 / 6, we need to follow the order of operations (PEMDAS):

Multiply 3 and 4: 3 * 4 = 12
Divide 5 by 6: 5 / 6 = 0.83 (approximately)
Add 2 and 12: 2 + 12 = 14
Subtract 0.83 from 14: 14 - 0.83 = 13.17

Therefore, the final answer is approximately 13.17.

Bot image for Deepseek-V3-FW

Deepseek-V3-FW
Poe

Certainly! Let's solve the expression step by step following the order of operations (PEMDAS/BODMAS: Parentheses/Brackets, Exponents/Orders, Multiplication and Division (left to right), Addition and Subtraction (left to right)).
Expression:

2 + 3 * 4 – 5 / 6

Multiplication (3 * 4):
Multiply 3 and 4 → 12.
Now: 2 + 12 − 5 / 6.

Division (5 / 6):
Divide 5 by 6 → ~0.833 (repeating).
Now: 2 + 12 − 0.833.

Addition (2 + 12):
Add 2 and 12 → 14.
Now: 14 − 0.833.

Subtraction (14 − 0.833):
Subtract 0.833 from 14 → ~13.167.

Final Answer:

≈ 13.167

(Note: Exact value is 14−56=846−56=796≈13.16714−65=684−65=679≈13.167).

Steps are summarized:

Multiply 3 × 4 = 12.
Divide 5 ÷ 6 ≈ 0.833.
Add 2 + 12 = 14.
Subtract 14 − 0.833 ≈ 13.167.

AI *for* Testing

- AI for testing
  - Test case and data generation
  - “Value” based testing
- Sample of work
  - **Blogs**: https://www.perfecto.io/blog/ai-in-software-testing ; https://www.testingxperts.com/blog/AI-in-Software-Testing
  - **Papers**: Artificial Intelligence in Software Test Automation: A Systematic Literature Review, Anna Trudova, Michal Dolezel, Alena Buchalcevová, Published in ENASE 2020, https://www.semanticscholar.org/paper/Artificial-Intelligence-in-Software-Test-A-Review-Trudova-Dolezel/ccbe24b348194905edeca78477625500786e55d6;
    T. M. King, J. Arbon, D. Santiago, D. Adamo, W. Chin and R. Shanmugam, "AI for Testing Today and Tomorrow: Industry Perspectives," *2019 IEEE International Conference On Artificial Intelligence Testing (AITest)*, 2019, pp. 81-88, doi: 10.1109/AITest.2019.000-3.

Mapping AI Methods to Test Activities

Table 4: Mapping AI techniques and testing activities (x = technique is applicable).

AI technique \ Testing activity	Publications identifier	Test case generation	Test oracle generation	Test execution	Test data generation	Test results reporting	Test repair	Test case selection	Flaky test prediction	Test order generation
Non-maximum suppression method (NMS)	R32					x				
SIFT, FAST, and FNCC algorithms	R24, R37						x			
Contour detection, OCR	R37						x			
Bayesian Network	R7, R28	x						x	x	
Particle swarm optimization (PSO)	R18							x		
Hybrid genetic algorithms	R14, R16	x			x					
Ant colony optimization (ACO)	R2, R8	x			x					
Artificial Neural Network (ANN)	R3, R11, R22, R23	x	x							
Graphplan algorithm	R9, R26	x	x							
Support vector machine (SVM)	R36, R38	x	x							
AdaBoostM1 and Incremental Reduced Error Pruning (IREP) algorithms	R33		x							
Convolutional Neural Networks (CNN)	R29		x		x					
Template-matching algorithm	R32, R35		x							
Decision tree algorithm (C4.5)	R4	x								
Markov model	R31	x								
MF-IPP (Multiple Fact Files Interference Progression Planner)	R15	x								
Algorithm from NLP field	R25	x								
Q-learning	R12, R17, R30	x								
Recurrent neural network (RNN)	R13				x					
L*	R39			x	x					
Fuzzing algorithm	R20			x						
k-means	R21			x						
KStar classifier	R19			x						
Heuristics algorithms	R27									x

Testing *for* AI

- Papers
  - A. Aggarwal, S. Shaikh, S. Hans, S. Haldar, R. Ananthanarayanan and D. Saha, "Testing Framework for Black-box AI Models," *2021 IEEE/ACM 43rd International Conference on Software Engineering: Companion Proceedings (ICSE-Companion)*, 2021, pp. 81-84, doi: 10.1109/ICSE-Companion52605.2021.00041. **Video**: https://youtu.be/984UCU17YZI
  - Machine Learning Testing: Survey, Landscapes and Horizons, Jie M. Zhang, Mark Harman, Lei Ma, Yang Liu, https://arxiv.org/abs/1906.10742, 2019
  - Software Engineering for AI-Based Systems: A Survey, Silverio Martínez-Fernández, Justus Bogner, Xavier Franch, Marc Oriol, Julien Siebert, Adam Trendowicz, Anna Maria Vollmer, Stefan Wagner, https://arxiv.org/abs/2105.01984, 2021

What is AI Being Tested For?

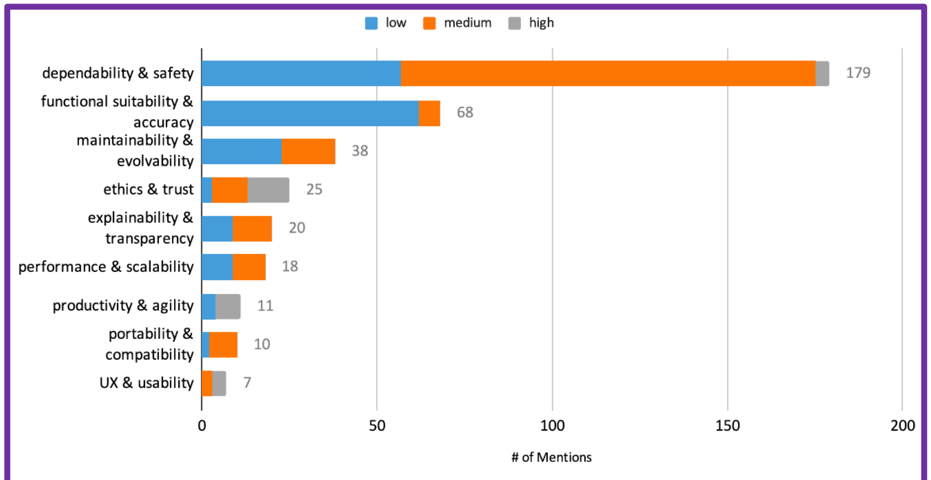


Fig. 16. Frequency of quality goals per thematic cluster and level of abstraction.

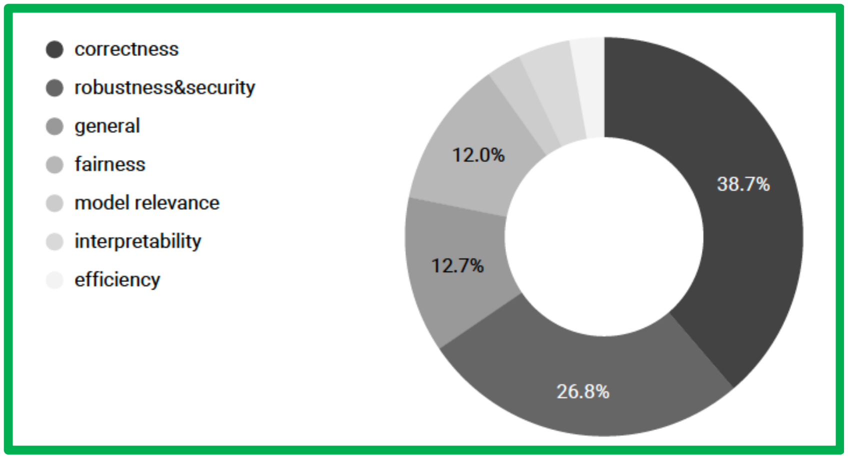


Figure Source: Software Engineering for AI-Based Systems: A Survey, Silverio Martínez-Fernández, Justus Bogner, Xavier Franch, Marc Oriol, Julien Siebert, Adam Trendowicz, Anna Maria Vollmer, Stefan Wagner, https://arxiv.org/abs/2105.01984, 2021

Figure Source: Machine Learning Testing: Survey, Landscapes and Horizons, Jie M. Zhang, Mark Harman, Lei Ma, Yang Liu, https://arxiv.org/abs/1906.10742, 2019

Figure Source: Machine Learning Testing: Survey, Landscapes and Horizons, Jie M. Zhang, Mark Harman, Lei Ma, Yang Liu, https://arxiv.org/abs/1906.10742, 2019

ML Testing

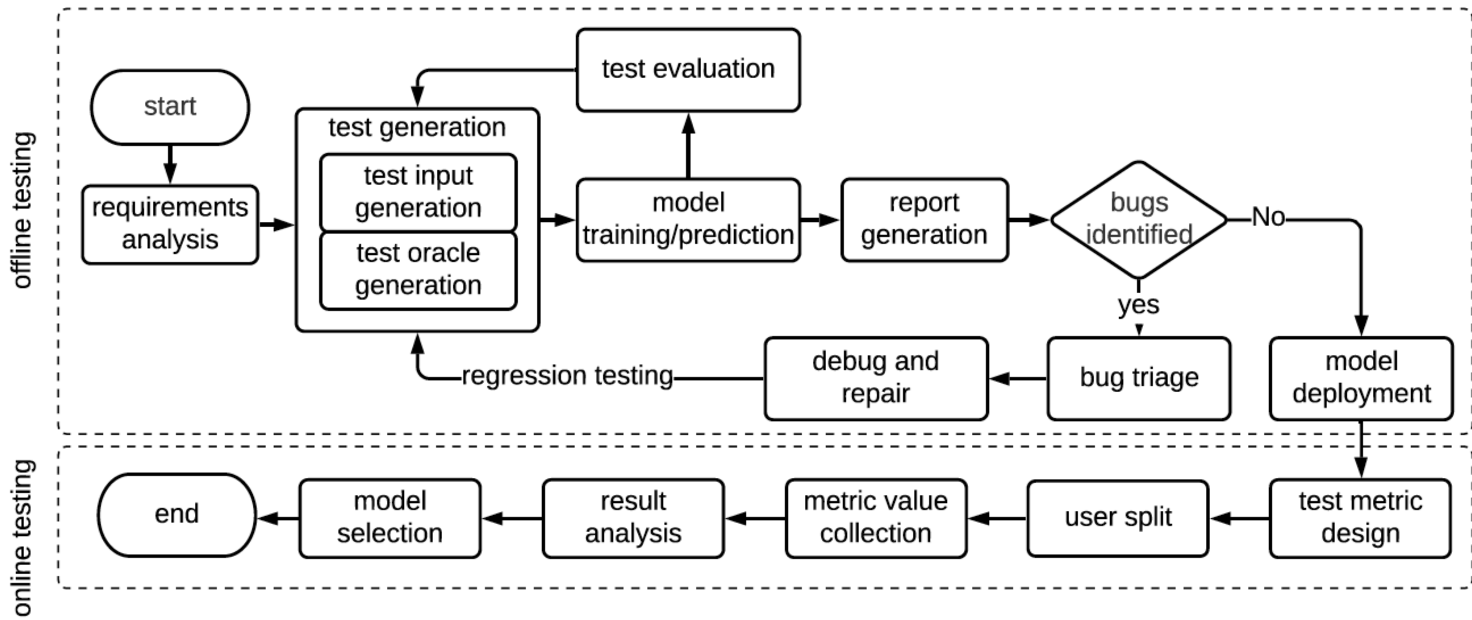


Figure 5: Idealised Workflow of ML testing

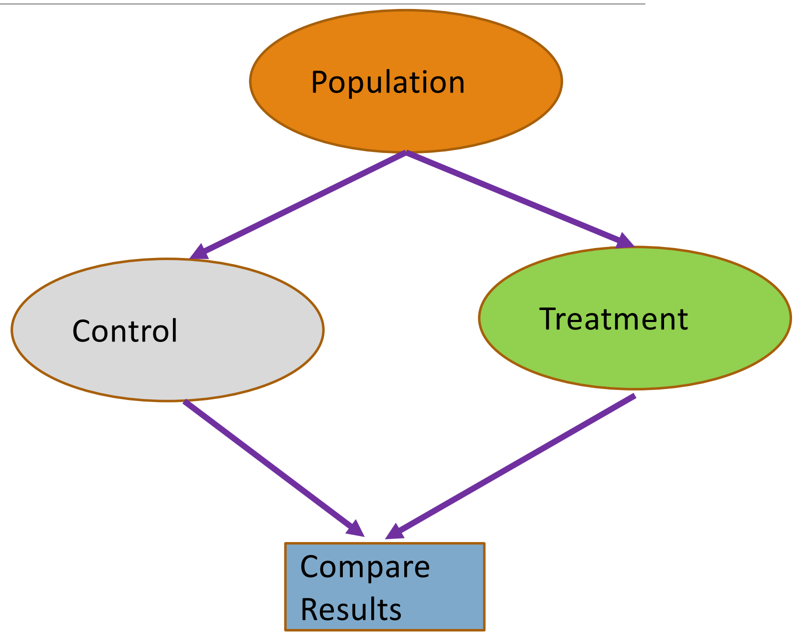
Causal Evaluation

Randomized Control Trial

- The population is ***randomly*** divided in two groups
- Control group gets placebo (nothing changes)
- Treatment group gets actual benefit that is being tested
- The difference in outcomes from the two groups are checked for statistical significance

Notes:

- population has to be large
- the split has to be random
- difference in treatment has to be ethical
- Considered gold standard in testing in critical domains like medicine, public policies



Example

Impact of mask on COVID cases

- https://theconversation.com/a-new-data-driven-model-shows-that-wearing-masks-saves-lives-and-the-earlier-you-start-the-better-149621

RCT for AI-based Systems

- Not very common for AI but increasing; against the prevalent culture
- Takes more effort than technical or business evaluation

Explanation for Text Classification

Explanation Taxonomy

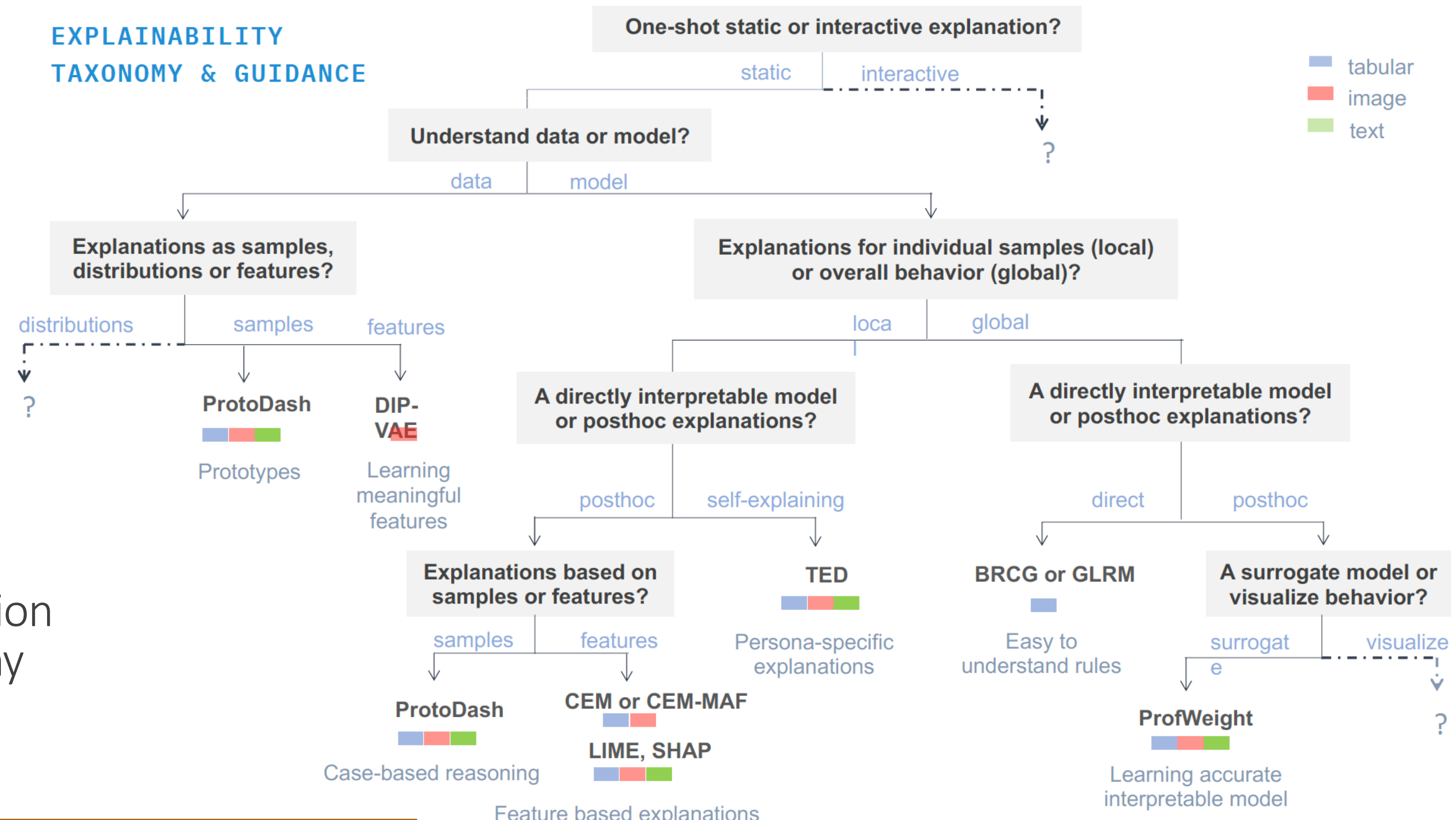


Figure Credit: Diptikalyan Saha and Vijay Arya, Oct 2021

Methods

- LIME:
  - Tools: LIME, InterpretML
- SHAP:
  - Tools: SHAP, ExplainerBoard

XAI for Text

LIME – Local Interpretable Model-Agnostic Explanations

Paper: "Why Should I Trust You?" Explaining the Predictions of Any Classifier, Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin, ACM's Conference on Knowledge Discovery and Data Mining, KDD2016

Blogs:

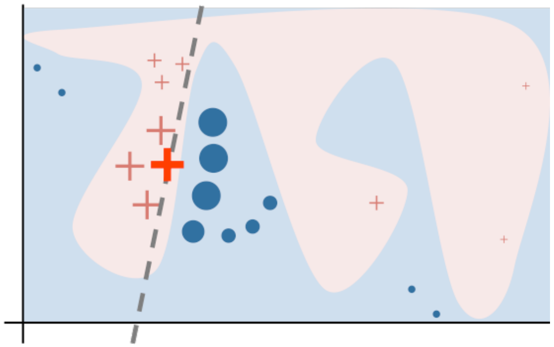
- https://homes.cs.washington.edu/~marcotcr/blog/lime/
- https://www.oreilly.com/content/introduction-to-local-interpretable-model-agnostic-explanations-lime/

Code: https://github.com/marcotcr/lime

Figures credit: Marco Tulio Ribeiro

LIME Key Idea

- Generate a local, linear explanation for any model
- How
  - Perturb near the neighborhood of a point of interest, X (**Local**)
  - Fit a linear function to the model's output (**Linear**)
  - Interpret coefficients of the linear function (**Explain**)
  - **Visualize**

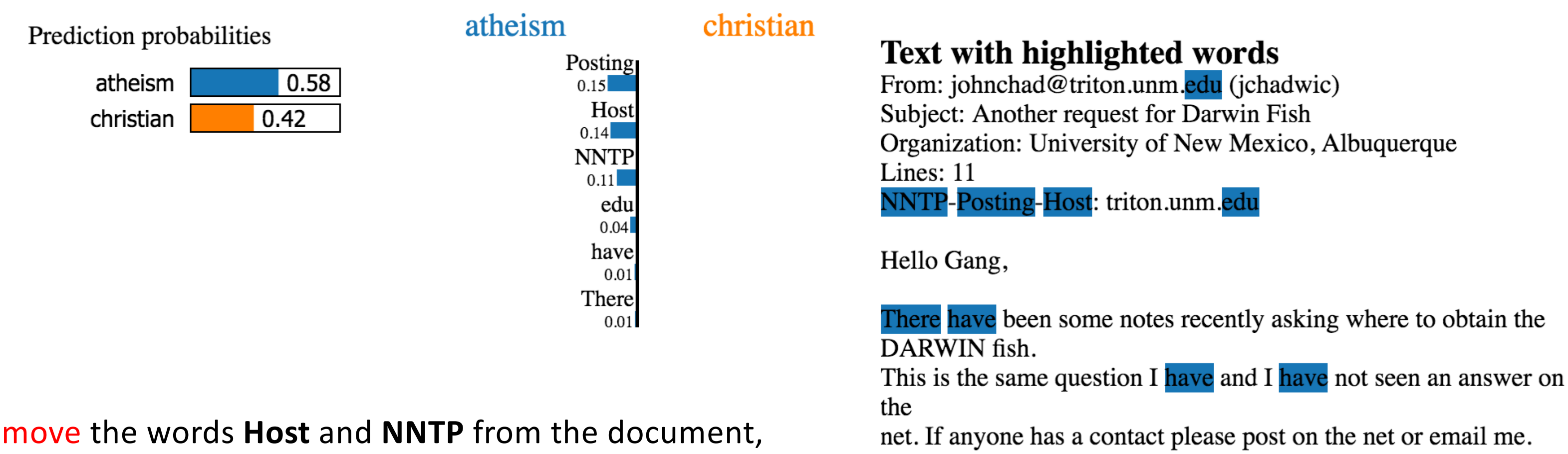


- Applicability
  - Any classification model!

LIME on Text

Question: Why is a classifier with >90% accuracy predicting based on ?

Task: classifying religious inclination from email text



"If we remove the words **Host** and **NNTP** from the document, we expect the classifier to predict **atheism** with probability 0.58 - 0.14 - 0.11 = 0.31"

Source: https://github.com/marcotcr/lime

Evaluation of Explanation Methods

- Text
  - Human-grounded Evaluations of Explanation Methods for Text Classification, Piyawat Lertvittayakumjorn, Francesca Toni, https://arxiv.org/abs/1908.11355, 2019

- Image
  - Opportunities and Challenges in Explainable Artificial Intelligence (XAI): A Survey, Arun Das, Paul Rad, https://arxiv.org/abs/2006.11371, 2020

- Many data types (image, text, audio, and sensory domains):
  - **How Can I Explain This to You? An Empirical Study of Deep Neural Network Explanation Methods,** *Jeya Vikranth Jeyakumar, Joseph Noor, Yu-Hsi Cheng, Luis Garcia, Mani Srivastava*, Advances in Neural Information Processing Systems 33 (NeurIPS 2020), https://proceedings.neurips.cc/paper/2020/hash/2c29d89cc56cdb191c60db2f0bae796b-Abstract.html

One Class, Many Explanations

An example from the Amazon dataset, Actual: Pos, Predicted: Pos, (Predicted scores: Pos 0.514, Neg 0.486):
"OK but not what I wanted: These would be ok but I didn't realize just how big they are. I wanted something I could actually cook with. They are a full 12" long. The handles didn't fit comfortably in my hand and the silicon tips are hard, not rubbery texture like I'd imagined. The tips open to about 6" between them. Hope this helps someone else know ..."

Method	**Top-3 evidence texts**	**Top-3 counter-evidence texts**
LIME (W)	comfortably / wanted / helps	not / else / someone
LRP (W)	are / not / 6	: / tips / open
LRP (N)	are hard , not / about 6" between / not what I wanted	. The tips open / : These would / in my hand and
Grad-CAM-Text (N)	comfortably in my hand / I wanted : These / . The tips open	not what I wanted / not rubbery texture like / Hope this helps someone
DTs (N)	imagined . The tips	'd imagined . / are . I wanted / would be ok

Table 3: Examples of evidence and counter-evidence texts generated by some of the explanation methods.

Image source: Human-grounded Evaluations of Explanation Methods for Text Classification, Piyawat Lertvittayakumjorn, Francesca Toni, https://arxiv.org/abs/1908.11355, 2019

Explanation Evaluation Tasks

	Task 1 (Section 3.1)	**Task 2** (Section 3.2)	**Task 3** (Section 3.3)
Assumption	Good explanations can reveal model behavior	Good explanations justify the predictions	Good explanations help humans investigate uncertain predictions
Model(s)	Two classifiers with different performance on a test dataset	One well-trained classifier	One well-trained classifier
Input text	A test example for which both classifiers predict the same class	A test example which the classifier predicts with high confidence ($\max_c \mathbf{p}_c > \tau_h$)	A test example which the classifier predicts with low confidence ($\max_c \mathbf{p}_c < \tau_l$)
Information displayed	1. The input text 2. The predicted class 3. (Highlighted) top-m evidence texts of each model	1. Top-m evidence texts	1. The predicted class 2. The predicted probability $\mathbf{p}$ 3. Top-m evidence and top-m counter-evidence texts
Human task	Select the more reasonable model and state if they are confident or not	Select the most likely class of the document which contains the evidence texts and state if they are confident or not	Select the most likely class of the input text and state if they are confident or not
Scores to the explanation method	(-)1.0: (In)correct, confident (-)0.5: (In)correct, unconfident 0.0: No preference	(-)1.0: (In)correct, confident (-)0.5: (In)correct, unconfident 0.0: No preference	(-)1.0: (In)correct, confident (-)0.5: (In)correct, unconfident

Table 1: A summary of the proposed human-grounded evaluation tasks.

Image source: Human-grounded Evaluations of Explanation Methods for Text Classification, Piyawat Lertvittayakumjorn, Francesca Toni, https://arxiv.org/abs/1908.11355, 2019

Review Example in Code

Github: https://github.com/biplav-s/course-tai-s25/blob/main/sample-code/LIME(Text)%20Classification.ipynb

InterpretML

- **Details**: https://github.com/interpretml/interpret
  - Whitebox (Glassbox) models: change learning code to introduce explainability support
  - Blackbox models: don't change learning code

Interpretability Technique	Type
Explainable Boosting	**glassbox model**
APLR	glassbox model
Decision Tree	glassbox model
Decision Rule List	glassbox model
Linear/Logistic Regression	glassbox model
SHAP Kernel Explainer	blackbox explainer
LIME	**blackbox explainer**
Morris Sensitivity Analysis	blackbox explainer
Partial Dependence	blackbox explainer

Interpret (Text)

See: https://github.com/interpretml/interpret-text

Explainers for sophisticated classifiers:

- Classical Text Explainer - (Default: Bag-of-words with Logistic Regression)
- Unified Information Explainer
- Introspective Rationale Explainer
- Likelihood Explainer
- Sentence Embedder Explainer
- Hierarchical Explainer

Classification - Multimodal

- Data is multimodal, i.e., two or more models.

- Example: financial (stock) prediction
  - In the domain of finance, data can be both numeric (stock price) and textual data (news).
  - Sample exercises:
    - https://github.com/Joeyipp/predict-stock-trends-news [Joey Yipp, CSCE 771, 2020]
    - https://www.kaggle.com/code/shtrausslearning/news-sentiment-based-trading-strategy
- Example: emergency room delay prediction
  - Electronic Health Record can have both numeric (pulse, heart rate) and textual data (clinical notes)
  - Reference: **Utilizing Predictive Analysis to Aid Emergency Medical Services**
    - https://link.springer.com/chapter/10.1007/978-3-030-93080-6_17

Explanation in Neural Network

- Visualize neural network activation for explanation
  - Illustration via code (Credit: Kausik Lakkaraju) - https://colab.research.google.com/drive/1ABWbuQ09qLJzITkKtYrMGUNCw-inQPDn?usp=sharing
- Mechanistic explanation
  - Mechanistic Interpretability for AI Safety -- A Review, Leonard Bereska, Efstratios Gavves, https://arxiv.org/abs/2404.14082
  - Illustrative code samples:
    - https://www.neelnanda.io/mechanistic-interpretability/quickstart
    - https://transformerlensorg.github.io/TransformerLens/content/getting_started_mech_interp.html

Explanation in LLMs

- Setting: one prompts a LLM and gets an output. How does one explain the result?
- Ideas
  - Prompt again with variation; analyze output, infer
    - Question: how to create variations?
    - Question: how to analyze outputs?
    - Question: how to infer about LLMs understanding?
  - Prompt another LLM, analyze output, infer
    - Question: which other LLM(s)
    - …
- Prompting methods
  - https://www.promptingguide.ai/introduction
  - See reading list on LLMs (https://github.com/biplav-s/course-tai-s25/blob/main/reading-list/Readme-LLMs.md)

Prompting of LLMs

A prompt contains any of the following elements:

- **Instruction** - a specific task or instruction you want the model to perform
- **Context** - external information or additional context that can steer the model to better responses
- **Input Data** - the input or question that we are interested to find a response for
- **Output Indicator** - the type or format of the output.

Credit:
https://www.promptingguide.ai/introduction/elements

Advanced reading:
The Prompt Report: A Systematic Survey of Prompting Techniques (June 2024, Feb 2026)
https://arxiv.org/pdf/2406.06608

Exercise

- Calculate 2 + 3 * 4 – 5 / 6 and express steps in English
- Explain which LLMs do correctly and how did you infer that

Llama-3-70b-Groq, vis Poe

I'd be happy to help with the calculation.

To evaluate the expression 2 + 3 * 4 – 5 / 6, we need to follow the order of operations (PEMDAS):

Multiply 3 and 4: 3 * 4 = 12
Divide 5 by 6: 5 / 6 = 0.83 (approximately)
Add 2 and 12: 2 + 12 = 14
Subtract 0.83 from 14: 14 - 0.83 = 13.17

Therefore, the final answer is approximately 13.17.

Bot image for Deepseek-V3-FW

Deepseek-V3-FW
Poe

Certainly! Let's solve the expression step by step following the order of operations (PEMDAS/BODMAS: Parentheses/Brackets, Exponents/Orders, Multiplication and Division (left to right), Addition and Subtraction (left to right)).
Expression:

2 + 3 * 4 – 5 / 6

Multiplication (3 * 4):
Multiply 3 and 4 → 12.
Now: 2 + 12 − 5 / 6.

Division (5 / 6):
Divide 5 by 6 → ~0.833 (repeating).
Now: 2 + 12 − 0.833.

Addition (2 + 12):
Add 2 and 12 → 14.
Now: 14 − 0.833.

Subtraction (14 − 0.833):
Subtract 0.833 from 14 → ~13.167.

Final Answer:

≈ 13.167

(Note: Exact value is 14−56=846−56=796≈13.16714−65=684−65=679≈13.167).

Steps are summarized:

Multiply 3 × 4 = 12.
Divide 5 ÷ 6 ≈ 0.833.
Add 2 + 12 = 14.
Subtract 14 − 0.833 ≈ 13.167.

Concluding Comments

- We looked at difference between regular software and AI systems, and the nature of testing practices
- AI and testing has two connotations
  - How AI has been used for software testing ?
  - How software testing for AI has to be done?
- We also looked at how to evaluate explanation methods
- There is need for randomized control trial to build lasting trust in AI

Certifying/ Rating with Text-based AI

Trust Issues – Mitigate via Ratings

- Communicate behavior via certificates / ratings (increase transparency)
  - But, let humans make decisions

Trust Dimensions
Competent
Reliable
Upholds human values
Allows human interaction

Transparency Through Documentation of Rating

Documentation about
- Outcome (e.g., Nutrition label, Electronic DataSheet, Factsheet)
- Process (e.g., SEI Capability Maturity Model, ISO 9001)

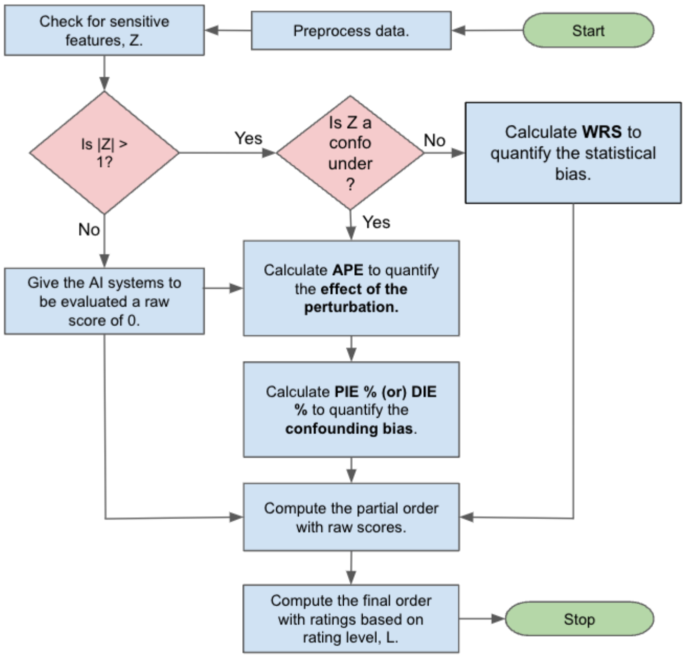
Documentation by
- Producer (e.g., Nutrition label)
- Consumer (e.g., Yelp rating)
- Independent 3rd Party (e.g., JD Powers, NHTSA car crash)

Reference: AboutML Project at PAI - https://www.partnershiponai.org/about-ml-get-involved/#read

ARC Tool and Rating

Tool: http://casy.cse.sc.edu/causal_rating

- See Sentiment Assessment Systems (SAS)



Explanation v/s Rating (Certification)

- Explanation
  - Focused on model behavior with respect to given input data
  - May give local or global explanation

- Certification
  - Gives general expected behavior
  - Global in nature; not specific to a particular data / input

- Complementary roles

Project Discussion

Project Milestones and Deliverables

- See course website for schedule and milestones: https://sites.google.com/site/biplavsrivastava/teaching/ai-csce-581-spring-2026-trusted-ai
  - Sprints: 1 (Crawl) : Feb 5-24, 2(Walk) : Feb 24-Mar31 and 3 (Run): Apr 2-21

  - Milestones – in-class presentations
    - Project discussion 1 – Feb 5
    - Project discussion 2 – March 5
    - **Project presentations 3 (final) – April 21 and 23**
    - **Project complete by April 30, 2026 in Github**
      - **Report, code, presentation (for completeness)**

Penalty

- Timelines of presentations [-10 points each miss; -30 total]
- Timelines of reports [-10 points]
- Code missing [-10 points]
- Testcase missing [-10 points]

Rubric for Evaluation of Course Project

Project

- Project plan along framework introduced (7 points)
- Challenging nature of project
- Actual achievement
- Report
- Sharing of code

Presentation

- Motivation
- Coverage of related work
- Results and significance
- Handling of questions

Penalty

- Timelines of presentations [-10 points each miss; -30 total]
- Timelines of reports [-10 points]
- Code missing [-10 points]
- Testcase missing [-10 points]

Course Project

- **Framework**
  1. (Problem) Think of a problem whose solution may benefit people (e.g., health, water, air, traffic, safety)
  2. (User) Consider how the primary user (e.g., patient, traveler) may be solving the problem today
  3. (AI Method) Think of what the solution will do to help the primary user
     1. Solution => ML task (e.g. classification), recommendation, text summarization, …
     2. Use a foundation model (e.g., LLM-based) solution as the baseline
  4. (Data) Explore the data for a solution to work
  5. (Reliability: Testing) Think of the evaluation metric we should employ to establish that the solution will works? (e.g., 20% reduction in patient deaths)
  6. (Holding Human Values) Discuss if there are fairness/bias, privacy issues?
  7. (Human-AI) Finally, elaborate how you will explain the primary user that your solution is trustable to be used by them

Project Discussion: What to Focus on ?

- Problem: you should care about it
- Data: should be available
- Method: you need to be comfortable with it. Have at least two – one serves as baseline
- Trust issue
  - Due to Users
    - Diverse demographics
    - Diverse abilities
    - Multiple human languages
  - Or other impacts
- What one does to mitigate trust issue

Project Discussion

1. Create a private Github repository called "CSCE581-Spring2025-<studentname>-Repo". Share with Instructor (biplav-s)
2. Create a folder called "Project". Inside, create a text file called "ProjectPlan.md" (or "ProjectPlan.txt") and have details by the next class (Jan 30, 2026)

1. Title:
2. Key idea: (2-3 lines)
3. Who will care when done:
4. Data need:
5. Methods:
6. Evaluation:
7. Users:
8. Trust issue:

Concluding Section

Week 11 (L23 and 24): Concluding Comments

- We looked at
  - L23: Explanations (Text)
  - L24: Assessment and Rating (Text)
  - Quiz 2

About Next Week – Lectures 25, 26

Lectures 25, 26

- Quiz 3 due
- Trust in Human-AI systems/ Chatbots
- Trust Standards and Laws, Privacy; Acceptable systems for the Real-world

17	Mar 17 (Tu)	**Invited Guest** –	W10 Quiz 2 - start
18	Mar 19 (Th)	AI - Unstructured (Text): Processing and Representation	
19	Mar 24 (Tu)	AI - Unstructured (Text): Representation, Common NLP Tasks, Large Language Models (LLMs)	[Week 11] Quiz 2- end
20	Mar 26 (Th)	Natural Languages/ Language Models and their Impact on AI	
21	Mar 31 (Tu)	AI - Unstructured (Text): Analysis – Supervised ML – Trust Issues	[Week 12]Project – Sprint 2 - end
22	Apr 2 (Th)	AI - Unstructured (Text): Analysis – Supervised ML – Mitigation Methods	Project – Sprint 3 - start
23	Apr 7 (Tu)	AI - Unstructured (Text): Analysis – Rating and Debiasing Methods	[Week 13] Quiz 3 - end
24	Apr 9 (Th)	Explanation Methods Trust: AI Testing	
25	Apr 14 (Tu)	Trust: Human-AI Collaboration	[Week 14] Quiz 3 - end
26	Apr 16 (Th)	Emerging Standards and Laws Trust: Data Privacy - Trusted AI for the Real World	
27	Apr 21 (Tu)	Project presentation	[Week 15] Project – Sprint 3 - end
28	Apr 23 (Th)	Project presentation	Last day of class
	Apr 28 (Tu)		Reading Day
29	April 30 (Th)	12:30pm – Final exam/ Overview	[Week 16]Optional, information shared